$\boxed{№11}$

$H_0: \xi \sim p_0(x) = 1 \{(0,1)\}$

$H_1: \xi \sim p_1(x) = \frac{e^{1-x}}{e-1} \{(0,1)\}$

а) $n=1$ с ур. знач-ти α $\boxed{G: x \le A}$

$$l(x) = \frac{L_1}{L_0} = \frac{p_1(x)}{p_0(x)} = \frac{e^{1-x}}{e-1} \ge c$$

$$P(x \le A \mid H_0) = \alpha$$

$$\int_0^A dx = \alpha \Rightarrow A = \alpha \Rightarrow G: x \le \alpha$$

$$W = P(x \le A \mid H_1) = \int_0^A \frac{e^{1-x}}{e-1} dx =$$

$$= \frac{e}{e-1} \cdot \frac{e^{-x}}{(-1)} \Big|_0^A = \frac{e}{e-1}(1-e^{-\alpha})$$

$\alpha_1 = \alpha$

$\alpha_2 = 1 - W$

б) $n=2$

$$l = \frac{L_1}{L_0} = \frac{\frac{e^{1-x_1}}{e-1} \cdot \frac{e^{1-x_2}}{e-1}}{1 \cdot 1} = \frac{e^{2-x_1-x_2}}{(e-1)^2} \ge c$$

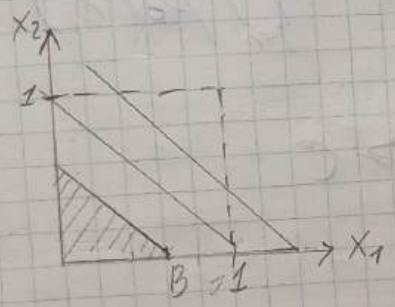
$G: x_1 + x_2 \le B$

$P(x_1 + x_2 \le B \mid H_0) = \alpha$

$$\iint\limits_{x_1+x_2\le B} dx_1\, dx_2 = \alpha$$

$$\frac{B^2}{2} = \alpha \Rightarrow B = \sqrt{2\alpha}$$

$$B \le 1$$



$$\sqrt{2\alpha} \le 1 \Rightarrow \alpha \le \frac{1}{2}$$

$G: x_1 + x_2 \le B = \sqrt{2\alpha}$

$$W = P(x_1 + x_2 \le B \mid H_1) = \iint\limits_{\substack{x_1+x_2\le B \\ 0\le x_1\le 1}} \frac{e^{1-x_1}\cdot e^{1-x_2}}{(e-1)^2}\, dx_1\, dx_2 =$$

$$= \frac{e^2}{(e-1)^2} \int_0^B dx_1 \int_0^{B-x_1} e^{-x_1}\cdot e^{-x_2}\, dx_2 =$$

$$= \frac{e}{(e-1)^2} \int_0^B e^{-x_1}\left(1 - e^{-B+x_1}\right) dx_1 =$$

$$= \frac{e^2}{(e-1)^2}\left[1 - e^{-B} - e^{-B}\cdot B\right], \text{ где } B = \sqrt{2\alpha}$$

$$\begin{cases} \alpha_1 = \alpha \\ \alpha_2 = 1 - W \end{cases}$$

в) Асимптотический критерий по выборке n

$$\ell(\vec{X_n}) = \frac{L_1(\vec{X_n})}{L_0(\vec{X_n})} = \frac{\prod_{i=1}^{n} p_1(x_i)}{\prod_{i=1}^{n} p_0(x_i)} =$$

$$= \prod_{i=1}^{n} \left(\frac{p_1(x_i)}{p_0(x_i)} \right) \geq C$$

$$\ln \ell = \sum_{i=1}^{n} \underbrace{\ln\left(\frac{p_1(x_i)}{p_0(x_i)} \right)}_{\eta_i} \geq \ln C$$

$$\Rightarrow \sum_{i=1}^{n} \ln p_1(x_i) - \sum_{i=1}^{n} \ln p_0(x_i) =$$

$$= \sum_{i=1}^{n} \ln\left(\frac{e^{1-x}}{e-1} \right) - 0 \geq \ln C$$

$$\sum_{i=1}^{n} (1 - x_i) - n \ln(e-1) \geq \ln C$$

$$G: \sum_{i=1}^{n} x_i \leq A \;\Big|\, :n \Rightarrow \bar{x} \leq B$$

$$P(\bar{x} \leq B \mid H_0) = \alpha$$

ЦПТ

$$\frac{\bar{x} - M\xi}{\sqrt{D\xi}}\sqrt{n} \rightsquigarrow N(0,1)$$

H_0:

$M\xi = \frac{1}{2}$; $P(\bar{x} \le B | H_0) =$

$D\xi = \frac{1}{12}$

$$= P\left(\frac{\bar{x} - \frac{1}{2}}{\sqrt{\frac{1}{12}}}\sqrt{n} \le \underbrace{\frac{B - \frac{1}{2}}{\sqrt{\frac{1}{12}}}\cdot\sqrt{n}}_{u_\alpha}\right)$$

$$B = \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}}$$

$$W = P(\bar{x} \le B | H_1)$$

$$H_1: M\xi = \int_0^1 x\, p_1(x)\,dx = \frac{e-2}{e-1}$$

$$M\xi^2 = \int_0^1 x^2 p_1(x)\,dx = \frac{2e-5}{e-1}$$

$$D\xi = \frac{e^2 - 3e + 1}{(e-1)^2}$$

$$P\left(\frac{\bar{x} - M\xi}{\sqrt{D\xi}}\cdot\sqrt{n} \le \frac{B - M\xi}{\sqrt{D\xi}}\sqrt{n}\right) =$$

$$= \int_{-\infty}^{\alpha} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}} dx = \frac{\frac{u_\alpha}{\sqrt{12}} + 0{,}082\sqrt{n}}{\sqrt{D\xi}}$$

$$\xrightarrow[n \to \infty]{} +\infty$$

$$\int \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}} \cdot dx \xrightarrow[n \to \infty]{} 1$$

$$\Longrightarrow \quad W \xrightarrow[n \to \infty]{} 1$$

$$\begin{cases} \alpha_1 = 1 \\ \alpha_2 = 1 - W \end{cases}$$

г) Построить оптимальный критерий по выборке объёма n с крит. обл. G: $x_{min} < C$ и ур-ем значимости α

$$\alpha_2 \to \min$$
$$\alpha_1 \le \alpha$$

$$\alpha_1 = P(x_{min} \le C / H_0) = 1 - (1 - f(C))^n \le \alpha$$

$$1 - (1 - C)^n \le \alpha \Longrightarrow \sqrt[n]{1 - \alpha} \ge C$$

$$\alpha_2 = P(x_{min} > c \mid H_1) =$$

$$= 1 - P(x_{min} \le c \mid H_1) =$$

$$= (1 - \Phi(c))^n =$$

$$= \left\{ \Phi(c) = \frac{e}{e-1}(1 - e^{-c}) \right\} =$$

$$= \left(1 - \frac{e}{e-1}(1 - e^{-c})\right)^n$$

$$\frac{d\alpha_2}{dc} = n\left(1 - \frac{e}{e-1}(1 - e^{-c})\right)^{n-1}\left(\frac{-e^{1-c}}{e-1}\right) = 0$$

$$\frac{e}{e-1}(1 - e^{-c}) = 1$$

$$\alpha_1 \le \alpha \Rightarrow 1 - \sqrt[n]{1-\alpha} \ge c$$

$$\boxed{e^{-c} = e^{-1} \Rightarrow c = 1}$$

$$1 - \alpha \in (0;1) \Rightarrow \sqrt[n]{1-\alpha} \in (0;1) \Rightarrow$$

$$\Rightarrow 1 - \sqrt[n]{1-\alpha} \in (0;1) \Rightarrow$$

$$\Rightarrow c > 1 - \sqrt[n]{1-\alpha} \text{ (усл. для } \alpha_2 \text{)}$$

$$\frac{d^2\alpha_2}{dc^2} = \underline{n(n-1)\left(1 - \frac{e}{e-1}(1 - e^{-c})\right)^{n-2}}.$$

$$\cdot\left(-\frac{e^{1-c}}{e-1}\right) + n\left(1-\frac{e}{e-1}(1-e^{-c})^{n-1}\cdot\left(\frac{e^{1-c}}{e-1}\right)\right)$$

$$> 0$$

$\Rightarrow c$ - m. min

$$c = 1-\sqrt[n]{1-\alpha} \Rightarrow c = 1-\sqrt[n]{1-\alpha}$$

$$\alpha_1 = \alpha$$

$$\Rightarrow G: x_{min} < 1-\sqrt[n]{1-\alpha}$$

$$W = P(x_{min} < 1-\sqrt[n]{1-\alpha} \mid H_1) =$$

$$= 1-(1-\Phi(c))^n =$$

$$= \left(1-\frac{e}{e-1}(1-e^{-c})\right)^n =$$

$$= 1-\left(1-\frac{e}{e-1}\left(1-e^{-1+\sqrt[n]{1-\alpha}}\right)\right)^n =$$

$$= \ldots = 1-\left(1-\frac{e}{e-1}\left(1-1-\frac{1}{n}\cdot\ln(1-\alpha)+\right.\right.$$

$$\left.\left.+O\left(\frac{1}{n}\right)\right)\right)^n =$$

$$= 1-\left(1+\frac{e}{e-1}\cdot\frac{1}{n}\ln(1-\alpha)+\right.$$

$$\left.+O\left(\frac{1}{n}\right)\right)^n \xrightarrow[n\to\infty]{} 1-$$

$$-e^{\frac{e\ln(1-\alpha)}{e-1}} =$$

$$= 1 - (1-\alpha)^{\frac{1}{r-1}} \neq 1$$

$\Rightarrow$ крит. не сост.

$\boxed{N12}$

$H_0: p_4 = \frac{8}{24}; p_3 = \frac{4}{24}; p_2 = p_1 = \frac{6}{24}$

$H_1: p_1 = p_2 = p_3 = p_4 = \frac{6}{24}$

$$\begin{cases} \alpha = 0{,}2 \\ n = 2 \end{cases} \qquad l(\vec{X_n}) = \frac{L_1(\vec{X_n})}{L_0(\vec{X_n})}$$

p_0:

H_0	1	2	3	4
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{24}$	$\frac{1}{12}$
2	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{24}$	$\frac{1}{12}$
3	$\frac{1}{24}$	$\frac{1}{24}$	$\frac{1}{36}$	$\frac{1}{18}$
4	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{9}$

p_1:

H_1	1	2	3	4
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
2	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
4	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$

l:

l	1	2	3	4
1	1	1	$\frac{3}{2}$	$\frac{3}{4}$
2	1	1	$\frac{3}{2}$	$\frac{3}{4}$
3	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{9}{4}$	$\frac{9}{8}$
4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{9}{8}$	$\frac{9}{16}$

$p_{ij} = p_i \cdot p_j$

$P(l \geq c | H_0) \leq 0{,}2$

Исходы C по убыванию:

Самые выгодные для „принятия H_1":

①	$(3,3)$
②	$(3,1)$ $(1,3)$ $(3,2)$ $(2,3)$
③	$(3,4)$ $(4,3)$

① $P_0(3,3) = \frac{1}{36}$

② $4 \cdot \frac{1}{24} = \frac{1}{6}$

③ $2 \cdot \frac{1}{18} = \frac{1}{6}$

$\Rightarrow \frac{1}{36} + \frac{1}{6} = \frac{7}{36} \simeq 0{,}1944 < \alpha = 0{,}2$

$\frac{7}{36} + \frac{1}{9} = \frac{11}{36} \simeq 0{,}3 > \alpha$

$\Rightarrow$ перебор

$\frac{9}{8} < C \leq \frac{3}{2} \Rightarrow C = \frac{3}{2}$

$G: \ C \geq \frac{3}{2}$ – крит. обл.

$W = 5 \cdot \frac{1}{16} = \underline{\underline{\frac{5}{16}}}$

$\alpha_1 = \frac{7}{36}$

$\alpha_2 = 1 - W = \frac{11}{16}$

$\boxed{№13}$ $\alpha = 0{,}05$; W – ?

Елисеев:

$n = 139$ $\sigma_{дл} = 5{,}722$ мм

$\sigma_{шир} = 4{,}612$ мм

Европа:

$m = 1000$ $\sigma_{дл} = 6{,}161$ мм

$\sigma_{шир.} = 3{,}055$ мм

$\xi \sim N(\theta_1, \theta_2^2)$ – длина

$\eta \sim N(\theta_1', (\theta_2')^2)$ – ширина

Длина:

1) $H_0: \theta_{2\,евр.}^2 = \theta_{2\,ел.}^2$

$H_1: \theta_{2\,евр}^2 \neq \theta_{2\,ел.}^2$

Ширина:

2) $H_0: (\theta_{2\,евр}')^2 = (\theta_{2\,ел}')^2$

$H_1: (\theta_{2\,евр}')^2 \neq (\theta_{2\,ел}')^2$

$S^2 = \frac{n}{n-1}\sigma^2$ $S_{дл.\,ел.}^2 = 32{,}9785$

$S_{дл.\,евр.}^2 = 37{,}9959$

$S_{шир.\,ел.}^2 = 21{,}4246$

$S_{шир.\,евр.}^2 = 25{,}5786$

$\tilde{\Delta}_{дл.} = \frac{S_{дл.\,ел.}^2}{S_{дл.\,евр.}^2} = 0{,}86795$

$\tilde{\Delta}_{шир.} = \frac{S_{шир.\,ел.}^2}{S_{шир.\,евр.}^2} = 0{,}8376$

$\tilde{\Delta} \sim f(n-1, m-1) = f(138, 999)$

$\Rightarrow G: \ \tilde{\Delta} \le f_{\frac{\alpha}{2}}(138, 999)$

или $\tilde{\Delta} \ge f_{1-\frac{\alpha}{2}}(138, 999)$

$f_{\frac{\alpha}{2}}(138, 999) = 0{,}76938$

$f_{1-\frac{\alpha}{2}}(138, 999) = 1{,}27{,}169$

$\Rightarrow \tilde{\Delta}_{\text{нач.}}$ и $\tilde{\Delta}_{\text{дл.}} \notin G$

$\Rightarrow$ Нет осн. отв. H_0 для двух случаев

$W = P(\Delta \in G \,/\, H_1) =$

$= P(\tilde{\Delta} \le f_{\frac{\alpha}{2}}(138, 999) \,/\, H_1) +$

$+ P(\tilde{\Delta} \ge f_{1-\frac{\alpha}{2}}(138, 999) \,/\, H_1) =$

$= P\left(\frac{S_{\text{н.}}^2}{S_{\text{вер.}}^2} \cdot \frac{\sigma_{\text{вер.}}^2}{\sigma_{\text{н.}}^2} \le f_{\frac{\alpha}{2}}(138, 999)\right) +$

$+ P\left(\frac{S_{\text{н.}}^2}{S_{\text{вер.}}^2} \cdot \frac{\sigma_{\text{вер.}}^2}{\sigma_{\text{н.}}^2} \ge f_{1-\frac{\alpha}{2}}(138, 999)\right) \Rightarrow$

$\Rightarrow W = \int_{f_{\frac{\alpha}{2}}(138, 999)/0}^{+\infty} q(t)\, dt +$

$+ \quad f_{1-\frac{\alpha}{2}}(138,999) / \Theta$

$\boxed{N14}$

$\xi \sim N(a, \sigma_x^2) \qquad \sigma_x^2 = 2$

$\eta \sim N(b, \sigma_y^2) \qquad \sigma_y^2 = 1$

$\vec{X_n} = [-1,14; -6,10; 2,42]$

$\vec{Y_m} = [-2,29; -2,91]$

$\alpha = 0,05 \qquad n = 3 \qquad m = 2$

$H_0: a = b \qquad$ по Ф Фишера $\qquad \frac{\bar{x}-a}{\sigma_x}\sqrt{n} \sim N(0,1)$

$H_1: a > b$

$\frac{\bar{y}-b}{\sigma_y}\sqrt{m} \sim N(0,1)$

$$\left.\begin{aligned} \bar{x} - a &\sim N\left(0, \frac{\sigma_x^2}{n}\right) \\ \bar{y} - b &\sim N\left(0, \frac{\sigma_y^2}{m}\right) \end{aligned}\right\} \Rightarrow \bar{x} - \bar{y} - (a-b) \sim N\left(0, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)$$

$$\Rightarrow \Delta = \frac{\bar{x} - \bar{y} - (a-b)}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0,1)$$

G:

$$\Delta \geq u_{1-\alpha}$$

$$\Delta = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0,1)$$

$$] a - b = (a)$$

$$\Rightarrow \bar{x} - \bar{y} \overset{H_1}{\sim} N\left(\theta, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)$$

$$\Rightarrow W = P(\hat{\Delta} \in G \mid H_1) =$$

$$= P\left(\hat{\Delta} \geq u_{1-\alpha}\left(\theta, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)\right)$$